AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

AI-Powered Customer Financial Health & Risk Intelligence

Project Domain

[Power BI with Copilot/Chatbot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. Problem Statement

Banks today collect vast amounts of customer transaction data, but they often struggle to convert this data into actionable business insights. A key challenge faced by financial institutions is the inability to clearly identify customers who are financially healthy versus those who are at risk.

This lack of visibility leads to poor lending decisions, increased chances of loan defaults, and missed opportunities for personalized financial services. Additionally, traditional reporting systems are static and do not provide real-time or predictive insights.

With the rise of AI tools such as Microsoft Copilot, there is an opportunity to enhance data analysis by generating automated insights and recommendations. However, these tools are underutilized due to lack of structured implementation.

This project aims to bridge this gap by analyzing customer transaction data to assess financial health, identify risk patterns, and provide actionable insights for better decision-making.

4. Objectives

- To document the workflow and host the project on GitHub with a demonstration video.
- To analyze customer transaction data to understand income, expenses, and financial behavior
- To calculate key financial health indicators such as Savings Rate, Spending Ratio, and Debt Burden
- To segment customers into categories such as Healthy, Moderate, and Risk.
- To identify high-risk customers for potential loan default
- To build an interactive Power BI dashboard for financial analysis
- To use Microsoft Copilot for generating AI-driven insights and recommendations

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	• Import banking dataset from Kaggle • Understand key columns such as transaction amount, category, account balance, and customer ID.
2	**Data Cleaning & Preprocessing**	Use Power Query to remove duplicates and fix incorrect data types. Develop "Role-based" AI prompts to help categorize unstructured transaction descriptions.
3	**Analysis & AI Integration**	Calculate KPIs such as: • *Savings Rate* • *Expense Ratio* • *Debt-to-Income Ratio* Use Copilot to generate summaries of customer financial behavior.
4	**Visualization / Output**	Build Power BI dashboard with: • *Financial Health Overview* • *Risk Segmentation* • *Spending Analysis* Include filters and drill-down capabilities.
5	**Documentation & Submission**	Finalize the report, record a walkthrough video of the dashboard filters, and push all code/files to GitHub.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable

- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- From Kaggle
- Link - https://www.kaggle.com/datasets/valakhorasani/bank-transaction-dataset-for-fraud-detection

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

- A clean, validated dataset with all preprocessing steps documented and a before/after comparison.
- An interactive dashboard or structured analytical report that clearly communicates key insights from the data.
- Demonstrated use of an AI tool (Microsoft Copilot or AI Chatbot) to generate insights, summaries, or content outputs, with prompts and results documented.
- A GitHub repository containing all project files, a completed fellowship project report, and a video demonstration of the working output.
- The project will generate a financial health score for each customer based on income, expenses, and financial behavior. This will help banks quickly understand which customers are financially stable and which are vulnerable

- Customers showing signs of financial stress (high expenses, low savings, irregular patterns) will be identified. This allows banks to reduce loan default risk and make more informed lending decisions.
- Customers will be grouped into meaningful categories such as healthy, moderate, and high-risk. This segmentation will help banks design targeted strategies for retention, cross-selling, and financial planning.
- The analysis will highlight key spending trends and patterns, helping banks understand how customers manage their money. This can reveal overspending habits and areas where financial discipline is lacking.
- Using Copilot, the project will generate automated insights and suggestions such as improving savings, reducing unnecessary expenses, or recommending suitable financial products. This enhances decision-making speed and efficiency.
- **Conclusion -** This project will transform raw banking data into meaningful business insights by focusing on customer financial health and risk identification. The integration of Power BI and Microsoft Copilot will enhance analytical capabilities and enable smarter, data-driven decision-making for financial institutions.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets